ChemComm

COMMUNICATION

View Article Online
View Journal | View Issue

Check for updates

Cite this: *Chem. Commun.*, 2021, **57**, 5394

Received 16th March 2021,
Accepted 27th April 2021

DOI: 10.1039/d1cc01422f

rsc.li/chemcomm

Published on 27 April 2021. Downloaded by Fudan University on 1/18/2022 2:08:04 AM.

# Engaging yne-allenones in tunable catalytic silane-mediated conjugate transfer reductions†

Xiao-Yan Qin,‡ Jia-Yin Wang,‡ Fang-Zhou Geng, Wen-Juan Hao,* Shu-Jiang Tu* and Bo Jiang*

**New tunable catalytic [2+2] cycloaddition/silane-mediated conjugate transfer reductions of yne-allenones have been developed, by which substituent-diverse cyclobutarenes with generally good yields were selectively synthesized by adjusting Fe–H and Cu–H catalytic systems. Use of the Fe–H system triggers 1,6-conjugate reduction to dihydrocyclobuta[*a*]naphthalen-4-ols whereas the Cu–H complex enables 1,4-conjugate reduction to cyclobuta[*a*]naphthalen-4(2*H*)-ones.**

As a class of highly strained carbocycles, cyclobutarenes show a wide spectrum of biological profiles including central nervous system depressants,[1a] and bradycardic[1b] and antitumoral[1c] agents. Specifically, cyclobutarenes, due to their structural rigidity and high intrinsic energy of the strained cyclobutene ring,[2] exhibit high reactivities and act as powerful and synthetically useful building blocks.[3] As a result, such molecules were widely applied in the assembly of various polycyclic scaffolds through inter- and intramolecular Diels–Alder reactions, with the key advance here involving the *in situ* generation of *ortho*-xylylene intermediates upon thermal activation.[4] The prominence of these cyclobutarenes has stimulated great efforts to establish effective methods for their syntheses, with these methods often including 1,4-elimination cycloaddition of functionalized arenes,[5] Parham cyclization,[6] photo-induced cycloadditions,[7] thermal extrusion reactions,[8] intramolecular addition of carbanions to benzynes,[9] [2+2+2] cycloadditions of 1,5-hexadiyne,[10] ring expansion of cycloproparenes,[11] [2+2] cycloadditions of allene precursors[12] and other methods.[13] However, most of the approaches so far developed rely on high temperatures (400–800 °C), strong bases (*n*-BuLi and $NaNH_2$), and multiple steps, or suffer from a narrow substrate scope and the need to carry out prefunctionalization of starting materials. Therefore, facile and practical protocols are still needed to synthesize cyclobutarenes and their derivatives from simple precursors and inexpensive catalysts.

Effective control of the chemoselectivity of the same reactants toward the structural diversity *via* chemodivergent strategies represents one of the extremely important tasks that could be realized by changing the catalysts and other reaction parameters.[14] Over the years, numerous chemodivergent methods have been developed to establish molecular diversity from readily available synthons.[15] Our group has presented several chemodivergent reactions toward polycyclic rings with bioactive importance, demonstrating high utility in skeletal diversity.[16] To continue our interest in the design of chemodivergent reactions by means of [2+2] cycloaddition of yne-allenones (Scheme 1a),[17] we questioned whether catalytic tunable silane-mediated conjugate reductions of yne-allenones could be achieved by combining diverse reductive systems, thereby levering a chemoselective entry to cyclobutarenes (Scheme 1a). With this concept in mind, we conducted an $Fe(acac)_3$-promoted reduction reaction between yne-allenones **1** and phenylsilane **2** in ethanol and found that the transformation proceeded through a [2+2] cycloaddition/1,6-conjugate reduction cascade, accessing 1,2-dihydrocyclobuta[*a*]naphthalen-4-ols **3** with generally good yields (Scheme 1b). Interestingly, exchanging the Fe–H complex for the Cu–H system resulted in a mechanistically different reductive pathway to give cyclobuta[*a*]naphthalene-4(2*H*)-ones **4** *via* a [2+2] cycloaddition/1,4-conjugate transfer reduction sequence (Scheme 1c). The current protocol represents the first combination of [2+2] cycloaddition with conjugate reductions starting from yne-allenones with silanes for the direct synthesis of new cyclobutarenes. Herein, we elaborate these catalytic chemodivergent reactions.

Our initial investigation commenced with the treatment of benzene-tethered yne-allenone **1a** with $PhSiH_3$ (**2a**) and $Fe(acac)_3$ in a 1 : 1 : 2 mole ratio at 60 °C under Ar conditions using ethanol (EtOH) as the reaction medium. The reduction reaction proceeded to generate the target product **3a** in 81%

*School of Chemistry & Materials Science, Jiangsu Normal University, Xuzhou, 221116, P. R. China. E-mail: wjhao@jsnu.edu.cn, laotu@jsnu.edu.cn, jiangchem@jsnu.edu.cn*

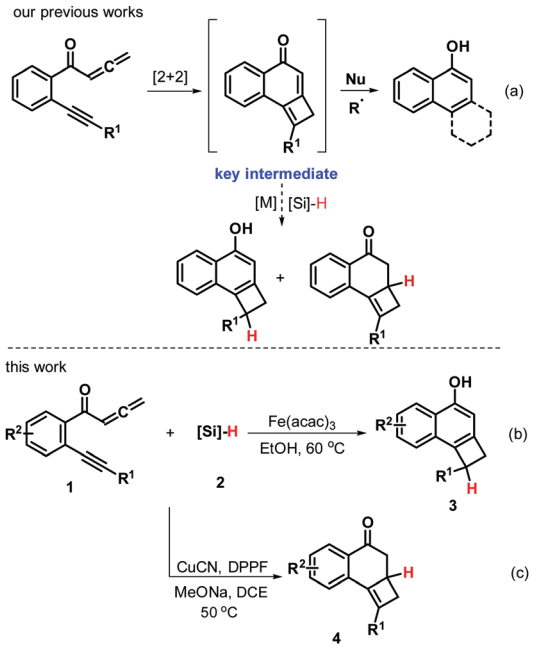
† Electronic supplementary information (ESI) available. See DOI: 10.1039/d1cc01422f

‡ These authors contributed equally.



This journal is © The Royal Society of Chemistry 2021

Published on 27 April 2021. Downloaded by Fudan University on 1/18/2022 2:08:04 AM.



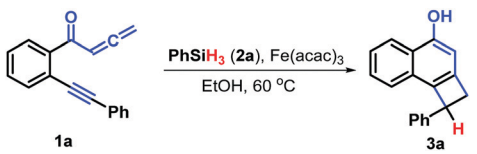

Scheme 1 Profiles for reductive hydrogenation of yne-allenones.

yield (Table 1, entry1). Lowering the dosage of $Fe(acac)_3$ was not beneficial for this transformation as lower yields of **3a** were observed (entries 2 and 3). In contrast, an increase in the amount of $Fe(acac)_3$ to 2.5 equivalents did not improve the efficiency of this reaction as well (entry 4). We then examined other transition-metal salts such as $Ru(acac)_3$ and $Co(acac)_2$ to evaluate this transformation, but both showed lower catalytic performances in comparison with $Fe(acac)_3$ (entry 1 *vs.* entries 5 and 6). The subsequent screening of the reaction media including methanol ($CH_3OH$) and tetrahydrofuran (THF) revealed that the reaction could work in these solvents, but gave lower yield than in EtOH (entry 1 *vs.* entries 7 and 8). Decreasing or increasing the reaction temperature showed no positive effect on the yield of **3a** (entries 9 and 10). Finally, several other silanes such as $Et_3SiH$, $Ph_3SiH$ and polymethylhydrosiloxane (PMHS) were screened, showing that all these reducing agents did not improve the yield of **3a** (entries 11–13).

With these optimized conditions in hand, we set out to systematically investigate the generality of this 1,6-conjugate reduction by examining a set of yne-allenones with different substitution patterns. As shown in Scheme 2a, the influence of substituents ($R^1$) in the arylalkynyl moiety of yne-allenones **1** was first evaluated. Substituents with diverse electronic and steric characteristics were well accommodated, and products **3b–3i** could be generated with comparable efficiency. Both electron-rich (methyl, **1b–1c**) and electron-poor (chloro, **1d–1e** bromo **1f**, trifluoromethyl **1g** and ester **1h**) groups at different positions can tolerate the reductive conditions well. Similarly, 2-thienyl analogue **1i** still showed a high reactivity profile, thus giving access to the corresponding 2-thienyl substituted product **3i** in 88% yield. Moreover, the electronic nature of the internal arene ring of yne-allenones was also briefly investigated. Various substituents such as methyl, methoxy, fluoro and chloro were introduced in the C4 or C5 position of the internal arene ring and these substrates were then reacted with phenylsilane under the standard conditions. As anticipated, the reactions proceeded readily, enabling Fe-mediated 1,6-conjugate reduction entry to approach the corresponding products **3j–3p** in 45–81% yields. The Fe-mediated reductive system can be easily amplified with a 4.0 mmol scale by carrying out the reaction of **1a** under the standard conditions (70%, Scheme 2b).

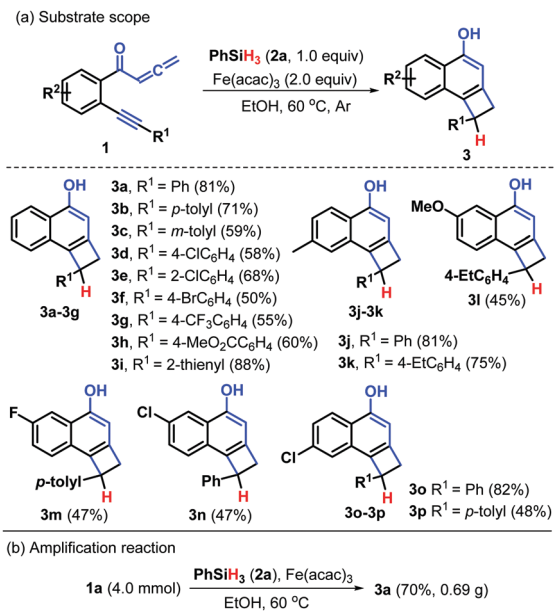
To expand the utility of this methodology, we turned to explore the feasibility of CuH-promoted conjugate reduction.

Table 1 Optimization of reaction conditions for **3a**[a]




| Entry | Variation from the standard conditions | Yield[b] (%) |
|---|---|---|
| 1 | None | 81 |
| 2 | 1.5 equiv. of $Fe(acac)_3$ | 64 |
| 3 | 1.0 equiv. of $Fe(acac)_3$ | 38 |
| 4 | 2.5 equiv. of $Fe(acac)_3$ | 67 |
| 5 | $Ru(acac)_3$ instead of $Fe(acac)_3$ | 62 |
| 6 | $Co(acac)_3$ instead of $Fe(acac)_3$ | 59 |
| 7 | $CH_3OH$ instead of EtOH | 58 |
| 8 | THF instead of EtOH | 42 |
| 9 | At 50 °C | 72 |
| 10 | At 80 °C | 60 |
| 11 | $Et_3SiH$ instead of $PhSiH_3$ | 45 |
| 12 | $Ph_3SiH$ instead of $PhSiH_3$ | 50 |
| 13 | PMHS instead of $PhSiH_3$ | 26 |

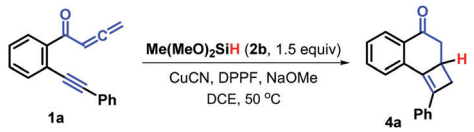
[a] Reaction conditions: **1a** (0.2 mmol), $PhSiH_3$ (**2a**, 0.2 mmol), $Fe(acac)_3$ (0.4 mmol), EtOH (2.5 mL); under Ar conditions; 1.5 h. [b] Isolated yield based on **1a**.




Scheme 2 Substrate scope for forming products **3**.

Published on 27 April 2021. Downloaded by Fudan University on 1/18/2022 2:08:04 AM.

Table 2 Optimization of reaction conditions for **4a**[a]




| Entry | Variation from the standard conditions | Yield[b] (%) |
|---|---|---|
| 1 | None | 90 |
| 2 | CuI instead of CuCN | Trace |
| 3 | CuBr instead of CuCN | 38 |
| 4 | $Cu(OAc)_2$ instead of CuCN | Trace |
| 5 | $Ph_3SiH$ instead of $Me(MeO)_2SiH$ | 48 |
| 6 | $Et_3SiH$ instead of $Me(MeO)_2SiH$ | NR |
| 7 | $PhSiH_3$ instead of $Me(MeO)_2SiH$ | NR |

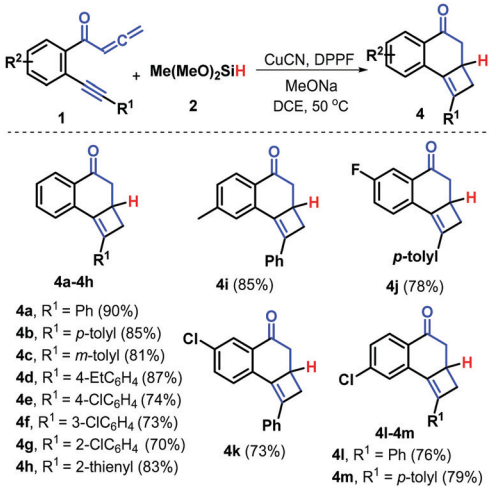
[a] Reaction conditions: **1a** (0.2 mmol), **2b** (1.5 equiv.), CuCN (10 mol%), DPPF (10 mol%), NaOMe (1.5 equiv.), DCE (2.5 mL), at 50 °C for 3 h, under Ar conditions. [b] Isolated yield based on **1a**.

By using CuCN and a 1,1′-bis(diphenylphosphino)ferrocene (DPPF) ligand as a catalytic system, the reaction of **1a** with methyldimethoxysilane (**2b**, $Me(MeO)_2SiH$) in a 1 : 1.5 mole ratio occurred smoothly in the presence of NaOMe in DCE at 50 °C under Ar conditions, and the unexpected dihydrocyclobuta[*a*]naphthalen-4(2*H*)-one **4a** was selectively afforded in 90% yield (Table 2, entry 1). The preliminary results prompted us to further search for the optimal conditions for accessing product **4a**. Several other copper salts including CuI, CuBr and $Cu(OAc)_2$ were tried for this transformation, but all these catalysts showed fairly lower catalytic abilities than CuCN (entries 2–4 *vs.* entry 1). Afterward, the effect of the types of silanes was probed, and the subsequent screening of other silanes such as $PhSiH_3$, $Et_3SiH$ and $Ph_3SiH$ indicated that all these silanes gave extremely unsatisfactory results (entries 5–7).

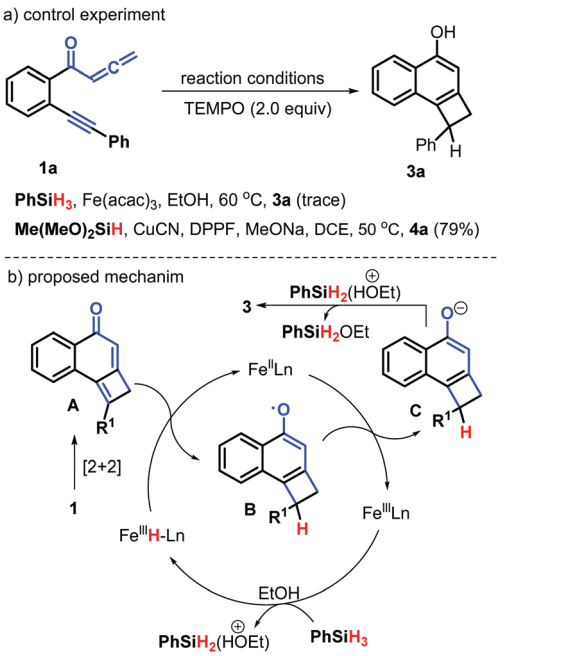
Having demonstrated the viability of engaging the yne-allenone **1a** in the chemodivergent reduction, we then investigated the substrate scope of this protocol (Scheme 3). At the outset, a variety of electron donating and withdrawing substituents (such as methyl, ethyl, and chloro) were installed into the phenyl ring relative to the alkynyl moiety of substrates **1** and in all the cases the desired products **4a–4g** were obtained with 70–90% yields. Moreover, 2-thienyl counterpart **1h** was a suitable surrogate for this transfer reduction (**4h**). Alternatively, yne-allenones bearing various substituents ($R^1$) including methyl, fluoro, and chloro at the C4 or C5 position of the internal arene ring were compatible with Cu–H catalytic conditions, furnishing dihydrocyclobuta[*a*]naphthalen-4(2*H*)-ones **4i–4m** with yields ranging from 73% to 85%.

It is well-known that the combination of $PhSiH_3$ and $Fe(acac)_3$ would provide Fe(III)-hydride species ($Fe^{III}Ln$–H), which could release a hydrogen radical. For this reason, 2.0 equivalents of 2,2,6,6-tetramethyl-1-piperidinyloxy (TEMPO) as a radical scavenger were subjected to the reaction of **1a** under the Fe(III)–H catalytic conditions (Scheme 4a). As expected, the progress of the reaction was completely suppressed, revealing that the reaction might include a hydrogen radical process. In addition, when 2.0 equivalents of TEMPO were added into the Cu-catalyzed system, the reaction could give product **4a** in 79% yield, indicating that this process did not involve the formation of a hydrogen radical (Scheme 4a).

On the basis of the above observations and literature reports,[18] a reasonable mechanism for product **3** is proposed in Scheme 4b. Initially, the Fe(III) complex reacts with phenylsilane in ethanol to give Fe(III)-hydride species, along with ethyl(phenylsilyl)oxonium. Then, Fe(III)-hydride species is trapped by cycloadduct **A**, generated from intramolecular [2+2] cycloaddition of yne-allenones **1**, thus leading to the formation of intermediate **B** through 1,6-hydrogen radical addition and the Fe(II) complex. Subsequently, single-electron reduction between **B** and the Fe(II) complex gives the stabilized naphthol anion **C** and regenerates the Fe(III) complex.



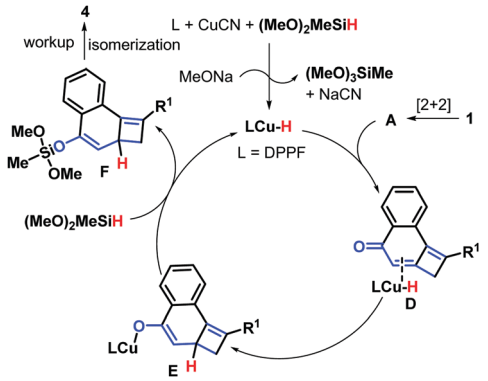

Scheme 3 Substrate scope for forming products **4**.




Scheme 4 Plausible reaction pathway for forming **3**.


This journal is © The Royal Society of Chemistry 2021

Published on 27 April 2021. Downloaded by Fudan University on 1/18/2022 2:08:04 AM.




Scheme 5 Plausible mechanism for the formation of **4**.

Intermediate **C** captures a proton from ethyl(phenylsilyl) oxonium to provide the final products **3**. In the formation of products **4** (Scheme 5), the active Cu–H species, *in situ*-generated from the reaction of CuCN with DPPF and hydrosilanes, is intercepted by cycloadduct **A** from yne-allenones **1** to form a π-complex **D**.[19] Then, delivery of the hydride to the β-carbon atom affords copper enolate intermediate **E**, which undergoes a σ-bond metathesis with hydrosilanes, thus leading to intermediate **F** and regenerating catalytically active Cu–H species. Intermediate **F** is converted into product **4** through desilicification by the workup process. The reason for this chemodivergent reaction depends on the difference in the catalytic abilities and modes between Fe–H and Cu–H complexes.

In conclusion, starting from easily available yne-allenones and hydrosilanes, new and tunable catalytic [2+2] cycloaddition/silane-mediated conjugate reductions were established for the first time, enabling the direct construction of substituent-diverse cyclobutarenes with generally good yields. In the former, Fe–H species enables a radical process to trigger 1,6-conjugate reduction to dihydrocyclobuta[*a*]naphthalen-4-ols with good yields whereas the latter generated new functionalized cyclobuta[*a*]naphthalen-4(2*H*)-ones *via* 1,4-conjugate reduction by adopting the Cu–H complex as the nucleophile. Both reductions show attractive advantages in terms of the efficient control of chemoselectivity from the same substrates and good compatibility with substituents, and thus opens new approaches for the divergent synthesis of structurally diverse cyclobuta[*a*]naphthalenes. Further investigation on their asymmetric versions is currently underway in our laboratories.

We are grateful for financial support from the National Natural Science Foundation of China (No. 21871112 and 21971090) and the Top-Notch Academic Programs Project of Jiangsu Higher Education Institution (TAPP).

## Conflicts of interest

There are no conflicts to declare.

## Notes and references

1 (*a*) J. A. Skorcz, *U.S. Pat.*, 3408391, 1968; (*b*) *WHO Drug Information*, 1996, vol. 10, p. 102; (*c*) S. Christophe, C. Kuehm-Caubere, P. Renard, B. Pfeiffer and P. Caubere, *Tetrahedron Lett.*, 1998, **39**, 9431.
2 (*a*) G. Mehta and S. Kotha, *Tetrahedron*, 2001, **57**, 625; (*b*) M. P. Cava and D. R. Napier, *J. Am. Chem. Soc.*, 1956, **78**, 500.
3 A. K. Sadana, R. K. Saini and W. E. Billups, *Chem. Rev.*, 2003, **103**, 1539.
4 (*a*) M. Iyoda, F. Sultana, S. Sasaki and H. Butenschon, *Tetrahedron Lett.*, 1995, **36**, 579; (*b*) J. G. Allen, M. F. Hentemann and S. J. Danishefsky, *J. Am. Chem. Soc.*, 2000, **122**, 571; (*c*) M. F. Hentemann, J. G. Allen and S. J. Danishefsky, *Angew. Chem., Int. Ed.*, 2000, **39**, 1937; (*d*) G. Ujaque, J. E. Norton and K. N. Houk, *J. Org. Chem.*, 2002, **67**, 7179; (*e*) K. Shishido, A. Yamashita, K. Hiroya, K. Fukumoto and T. Kametani, *Tetrahedron Lett.*, 1989, **30**, 111; (*f*) K. Shishido, A. Yamashita, K. Hiroya, K. Fukumoto and T. Kametani, *Tetrahedron*, 1989, **45**, 5791.
5 (*a*) M. K. Shepherd, *J. Chem. Soc.*, 1986, **17**, 1495; (*b*) B. D. Lenihan and H. Shechter, *J. Org. Chem.*, 1998, **63**, 2086; (*c*) B. D. Lenihan and H. Shechter, *Tetrahedron Lett.*, 1994, **35**, 7505; (*d*) H. Hart and R. W. Fish, *J. Am. Chem. Soc.*, 1961, **83**, 4460; (*e*) P. Schiess, S. Rutschmann and V. V. Toan, *Tetrahedron Lett.*, 1982, **23**, 3669; (*f*) P. Schiess, *Thermochim. Acta*, 1987, **112**, 31; (*g*) P. F. T. Schirch and V. Boekelheide, *J. Am. Chem. Soc.*, 1981, **103**, 6873.
6 (*a*) C. K. Bradsher and D. A. Hunt, *J. Org. Chem.*, 1981, **46**, 4608; (*b*) S. L. Buchwald, E. A. Lucas and J. C. Dewan, *J. Am. Chem. Soc.*, 1987, **109**, 439; (*c*) S. L. Buchwald, B. T. Watson, R. T. Lum and W. A. Nugent, *J. Am. Chem. Soc.*, 1987, **109**, 7137; (*d*) K. L. Dhawan, B. D. Gowland and T. Durst, *J. Org. Chem.*, 1980, **45**, 922; (*e*) E. Akgun, M. B. Glinski, K. L. Dhawan and T. Durst, *J. Org. Chem.*, 1981, **46**, 2730; (*f*) P. Beak and G. W. Selling, *J. Org. Chem.*, 1989, **54**, 5574.
7 (*a*) W. E. Parham, L. D. Jones and Y. A. Sayed, *J. Org. Chem.*, 1976, **41**, 1184; (*b*) D. C. Neckers and F. L. Wagenaar, *J. Org. Chem.*, 1981, **46**, 3939; (*c*) N. Hoffmann and J.-P. Pete, *Tetrahedron Lett.*, 1996, **37**, 2027.
8 (*a*) S. V. D'Andrea, J. P. Freeman and J. Szmuszkovicz, *J. Org. Chem.*, 1990, **55**, 4356–4358; (*b*) N. G. Andersen, S. P. Maddaford and B. A. Keay, *J. Org. Chem.*, 1996, **61**, 2885; (*c*) D. N. Hickman, T. W. Wallace and J. M. Wardleworth, *Tetrahedron Lett.*, 1991, **32**, 819.
9 J. F. Bunnett and J. A. Skorcz, *J. Org. Chem.*, 1962, **27**, 3836.
10 R. L. Funk and K. P. C. Vollhardt, *J. Am. Chem. Soc.*, 1980, **102**, 5245.
11 S. Kagabu and K. Saito, *Tetrahedron Lett.*, 1988, **29**, 675.
12 (*a*) J. E. Ezcurra and H. W. Moore, *Tetrahedron Lett.*, 1993, **34**, 6177; (*b*) J. Inanaga, Y. Sugimoto and T. Hanamoto, *Tetrahedron Lett.*, 1992, **33**, 703.
13 (*a*) P. J. Garratt and D. N. Nicolaides, *J. Org. Chem.*, 1974, **39**, 2222; (*b*) J. H. Markgraf, R. J. Katt, W. L. Scott and R. N. Shefrin, *J. Org. Chem.*, 1969, **34**, 4131; (*c*) M. P. Cava and D. R. Napier, *J. Am. Chem. Soc.*, 1957, **79**, 3606.
14 M. Wang, X. Zhang, Y.-X. Zhuang, Y.-H. Xu and T.-P. Loh, *J. Am. Chem. Soc.*, 2015, **137**, 1341.
15 V. M. Dembritsky, *Nat. Med.*, 2008, **62**, 1.
16 (*a*) J. Zhang, S.-Q. Shi, W.-J. Hao, G.-Y. Dong, S.-J. Tu and B. Jiang, *J. Org. Chem.*, 2021, DOI: 10.1021/acs.joc.0c02898; (*b*) D. Wang, S. Liu, X.-C. Lan, A. Paniagua, W.-J. Hao, G. Li, S.-J. Tu and B. Jiang, *Adv. Synth. Catal.*, 2017, **359**, 3186; (*c*) D.-F. Jiang, J.-Y. Hu, W.-J. Hao, S.-J. Tu and B. Jiang, *Org. Chem. Front.*, 2018, **5**, 189; (*d*) D. Wang, S.-C. Wang, W.-J. Hao, S.-J. Tu and B. Jiang, *Adv. Synth. Catal.*, 2020, **362**, 3416; (*e*) Z.-J. Shen, S.-C. Wang, W.-J. Hao, S.-Z. Yang, S.-J. Tu and B. Jiang, *Adv. Synth. Catal.*, 2019, **361**, 3837.
17 (*a*) B.-Z. Tang, W.-J. Hao, J.-Z. Li, S.-S. Zhu, S.-J. Tu and B. Jiang, *Chem. Commun.*, 2020, **56**, 7749; (*b*) S.-S. Zhu, J.-N. Zhou, Q.-L. Wu, W.-J. Hao, S.-J. Tu and B. Jiang, *Org. Chem. Front.*, 2020, **7**, 2975; (*c*) Q. Li, M. Li, S. Shi, X. Ji, C. He, B. Jiang and W. Hao, *Chin. J. Org. Chem.*, 2020, **40**, 384; (*d*) P. Zhou, J.-Y. Wang, T.-S. Zhang, G. Li, W.-J. Hao, S.-J. Tu and B. Jiang, *Chem. Commun.*, 2018, **54**, 164; (*e*) H.-K. Sha, F. Liu, J. Lu, Z.-Q. Liu, W.-J. Hao, J.-L. Tang, S.-J. Tu and B. Jiang, *Green Chem.*, 2018, **20**, 3476; (*f*) Y. He, T. Feng and X. Fan, *Org. Lett.*, 2019, **21**, 3918; (*g*) H.-K. Sha, T. Xu, F. Liu, B.-Z. Tang, W.-J. Hao, S.-J. Tu and B. Jiang, *Chem. Commun.*, 2018, **54**, 10415.
18 (*a*) J. Gui, C.-M. Pan and Y. Jin, *Science*, 2015, **348**, 886; (*b*) J. C. Lo, Y. Yabe and P. S. Baran, *J. Am. Chem. Soc.*, 2014, **136**, 1304; (*c*) H. T. Dao, C. Li and Q. Michaudel, *J. Am. Chem. Soc.*, 2015, **137**, 8046; (*d*) Q. Zhao, S.-J. Tu and B. Jiang, *Acta Chim. Sin.*, 2019, **77**, 927.
19 For selected reviews, see: (*a*) C. Deutsch, N. Krause and B. H. Lipshutz, *Chem. Rev.*, 2008, **108**, 2916; (*b*) S. Díez-González and S. P. Nolan, *Acc. Chem. Res.*, 2008, **41**, 349; (*c*) A. J. Jordan, G. Lalic and J. P. Sadighi, *Chem. Rev.*, 2016, **116**, 8318For selected examples: see ; (*d*) L. Rupnicki, A. Saxena and H. W. Lam, *J. Am. Chem. Soc.*, 2009, **131**, 10386; (*e*) D. Xiong, W. Zhou, Z. Lu, S. Zeng and J. Wang, *Chem. Commun.*, 2017, **53**, 6844; (*f*) Y. Zhou, J. S. Bandar, R. Y. Liu and S. L. Buchwald, *J. Am. Chem. Soc.*, 2018, **140**, 606.